Goalkeepers

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Player counts

In this project, there are a total of 60 goalkeepers considered, with

- 15 goalkeepers in 12 teams in the English FA Women's Super League
- 14 goalkeepers in 12 teams in the German Frauen Bundesliga
- 19 goalkeepers in 16 teams in the Spanish Liga F
- 12 goalkeepers in 10 teams in the Italian Serie A Women

Results

Results for 2 clusters

Breakdown of results by league

league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_2_cluster				
0	10	9	15	8
1	5	5	4	4

The split for two clusters is around 2 for 1 for all leagues, although Liga F is much more lopsided.

Breakdown of results by features

- Cluster 0 players on average had more share of their passes on the ground and from keepers' arms. Eventually this can align to a more modern style of ball playing goalkeepers
- On the other hand, cluster 1 players on average attempted longer passes including under pressure and had more share of passes from drop kicks. This can align to more traditional goalkeepers.

Eventually goalkeeper qualities such as saving or sweeping do not contribute to clustering results as much as passing qualities.

Results for 3 clusters

Breakdown of results by league

league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_3_cluster				
0	3	3	3	3
1	6	7	10	6
2	6	4	6	3

The split is more concentrated towards cluster 1 for all leagues.

Comparing results for 2 clusters against 3 clusters

kmeans_3_cluster	0	1	2
kmeans_2_cluster			
0	12	24	6
1	0	5	13

Most of k = 2 cluster 1 is retained in k = 3 cluster 2, while k = 3 is where both clusters break into, with more coming from k = 2 cluster 0.

Breakdown of results by features

- Cluster 0 goalkeepers stand out for high share of passes on the ground, passes from keepers' arms, and lower sweeping positions. These keepers can be read as ball playing keepers.
- While cluster 1 goalkeepers, in contrast, sweep at higher positions, while passing share stays in the middle of the other two clusters. These keepers can be interpreted as sweeper keepers.
- Cluster 2 goalkeepers are characterized by higher average pass lengths including under pressure. Similar to k = 2 cluster 1, these keepers are more traditional than the other two types.

Once again, goalkeeper positioning and saving by body parts do not contribute to clustering, although sweeping positioning does.

However, these results are not robust as number of components are added, so results are not highly confident.

Breakdown of results by players

Findings and radar plots for each goalkeeper are in their own files:

- Zecira Musovic - Chelsea
- Hannah Hampton - Chelsea
- Rebecca Leigh Spencer - Tottenham Hotspur
- Maria Korenciova - Como
- Mary Earps - Manchester United

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Appendix

Radar plots for 2 clusters goalkeepers' averages

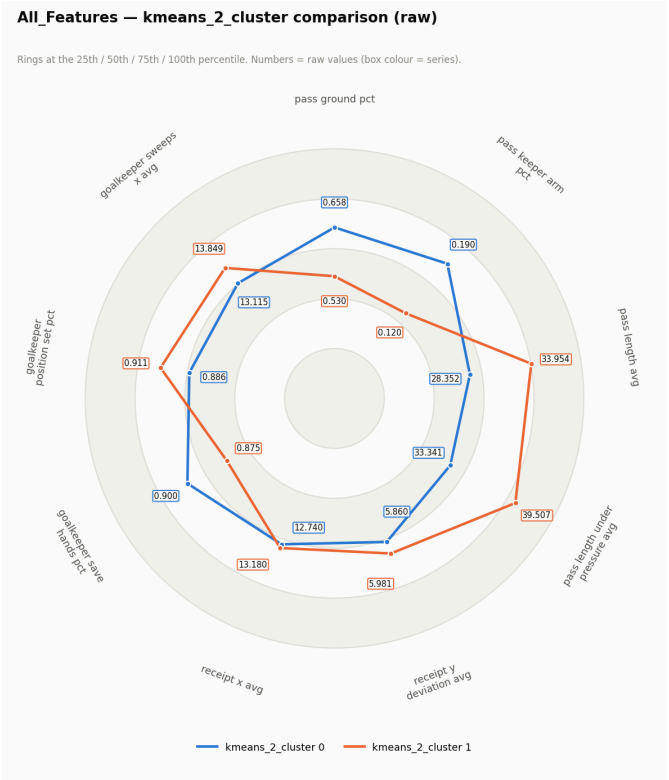


Figure 1: All Goalkeepers - Raw Statistics Comparison

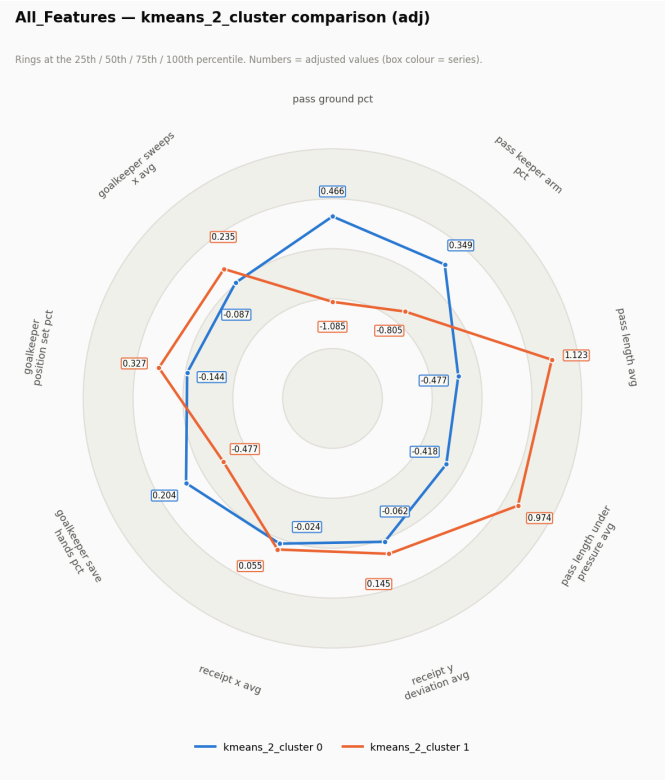


Figure 2: All Goalkeepers - Adjusted Statistics Comparison

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FA Women's Super League

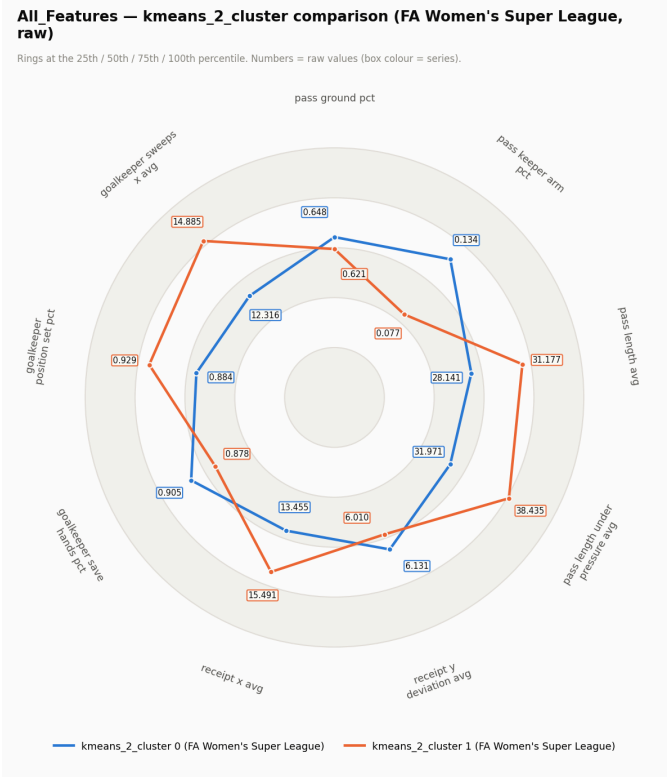


Figure 3: FA Women's Super League Goalkeepers — Raw Statistics Comparison

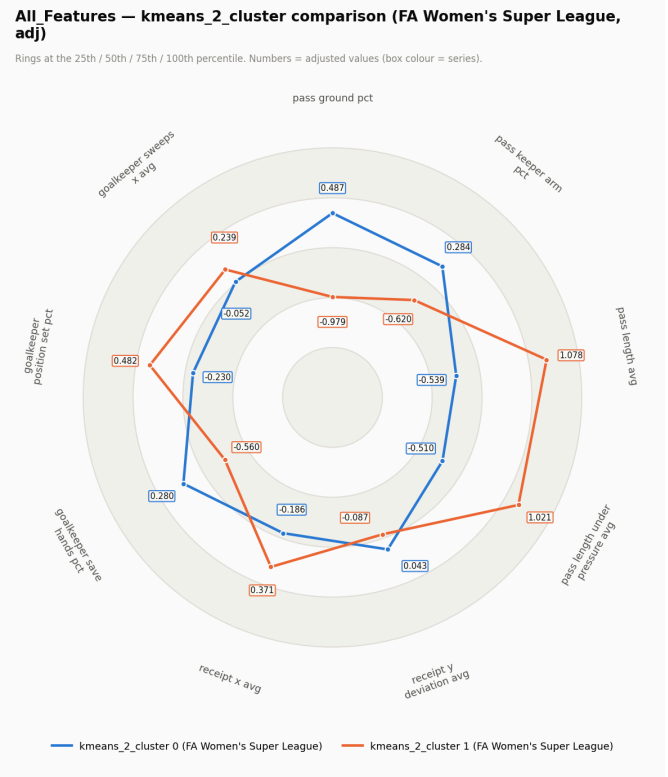


Figure 4: FA Women's Super League Goalkeepers — Adjusted Statistics Comparison

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Frauen Bundesliga

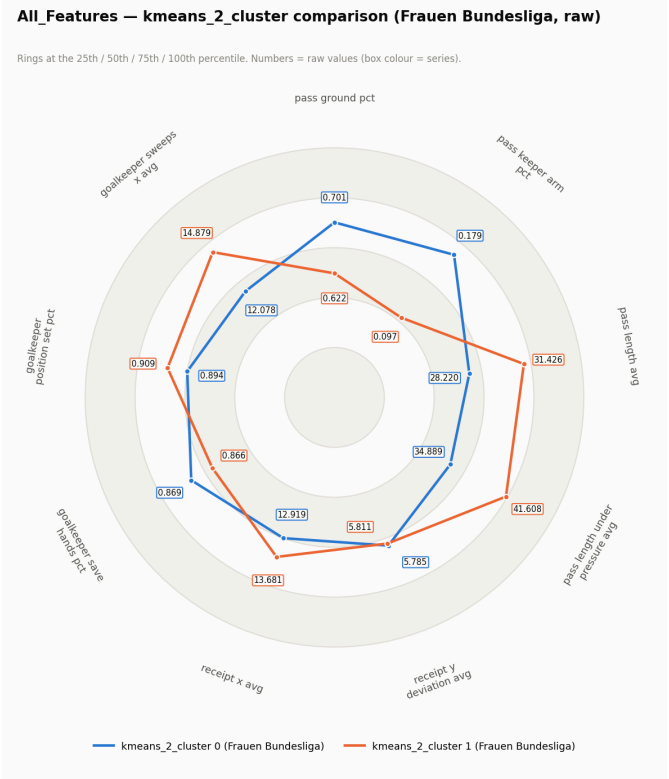


Figure 5: Frauen Bundesliga Goalkeepers — Raw Statistics Comparison

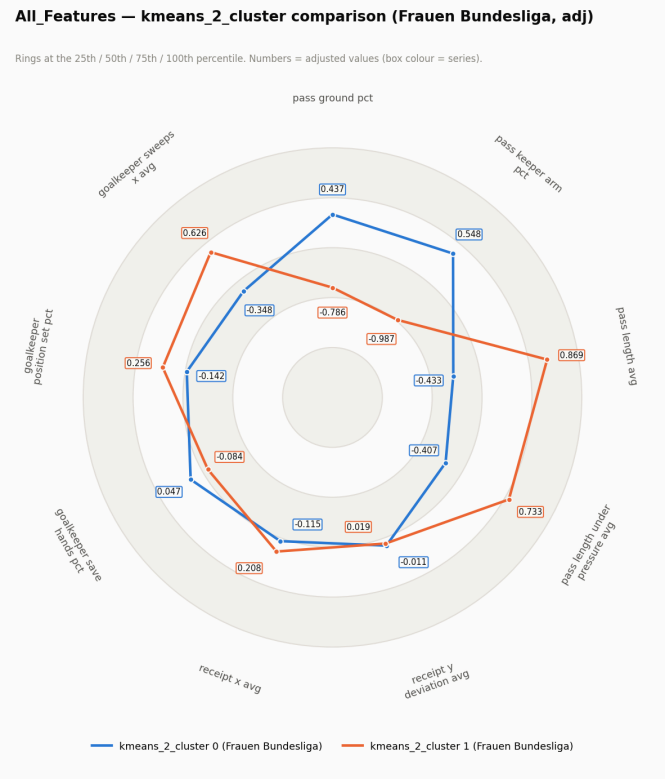


Figure 6: Frauen Bundesliga Goalkeepers — Adjusted Statistics Comparison

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Liga F

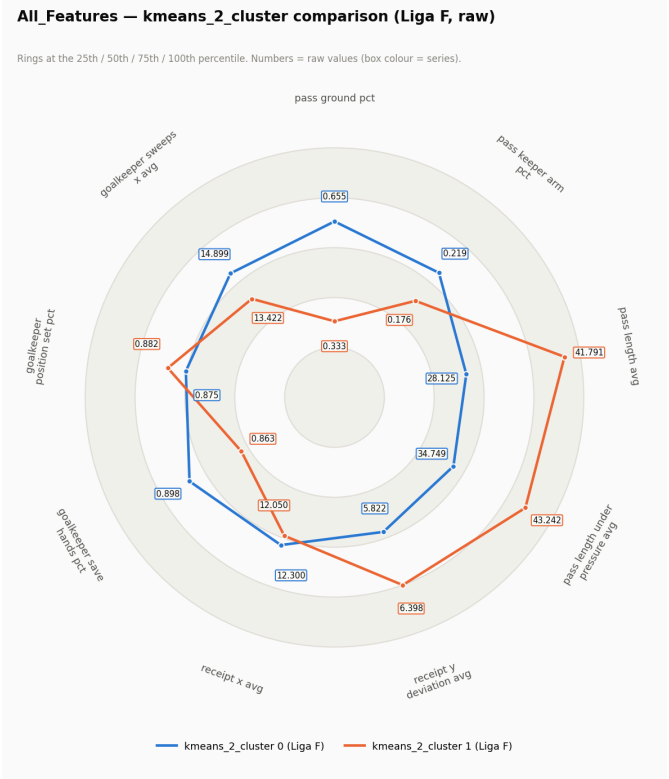


Figure 7: Liga F Goalkeepers — Raw Statistics Comparison

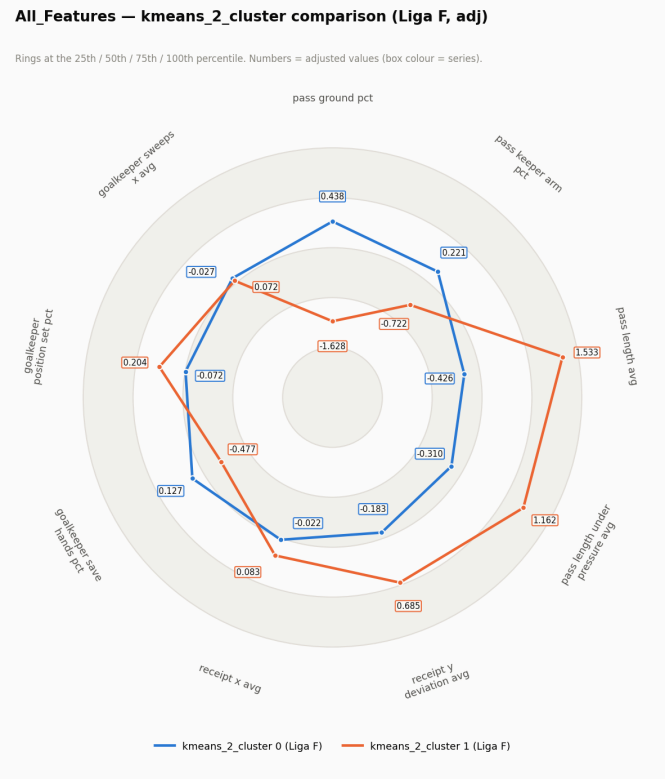


Figure 8: Liga F Goalkeepers — Adjusted Statistics Comparison

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Serie A Women

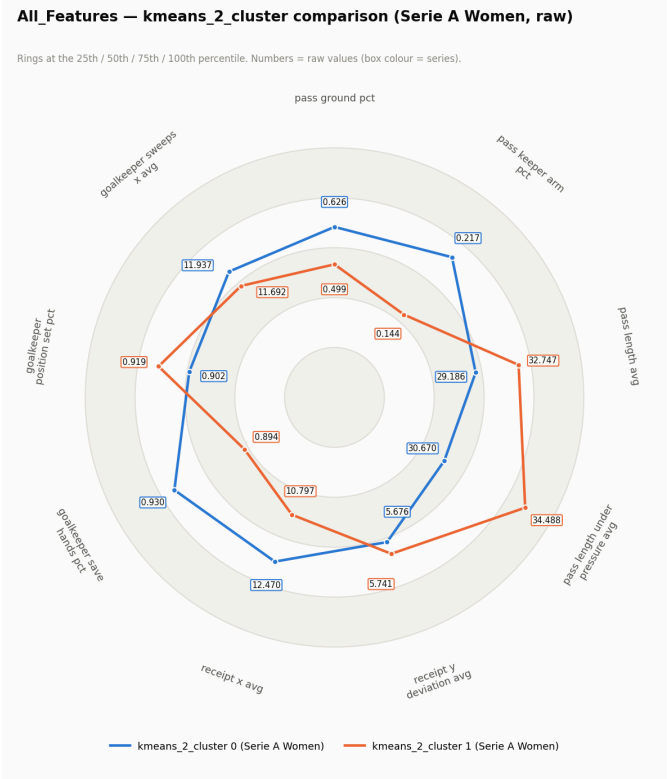


Figure 9: Serie A Women Goalkeepers — Raw Statistics Comparison

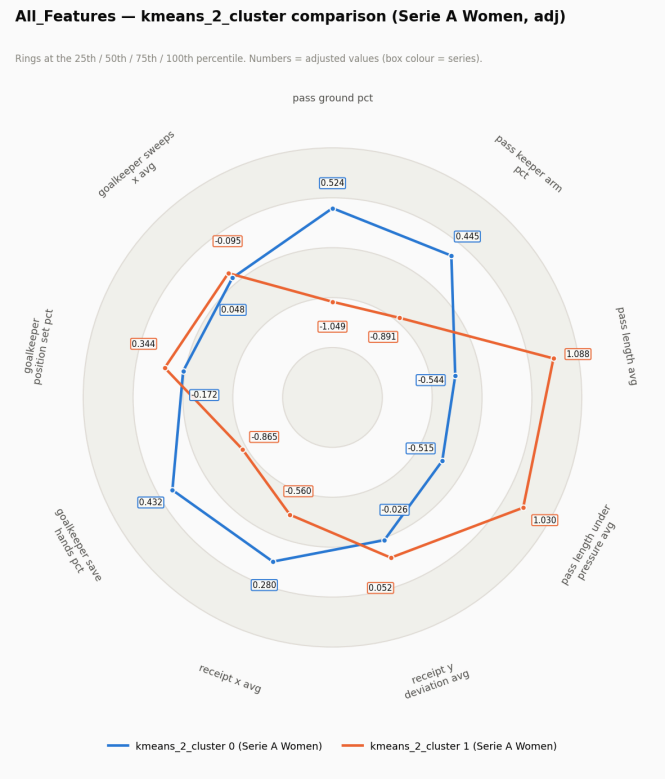


Figure 10: Serie A Women Goalkeepers — Adjusted Statistics Comparison

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Radar plots for 3 clusters goalkeepers' averages

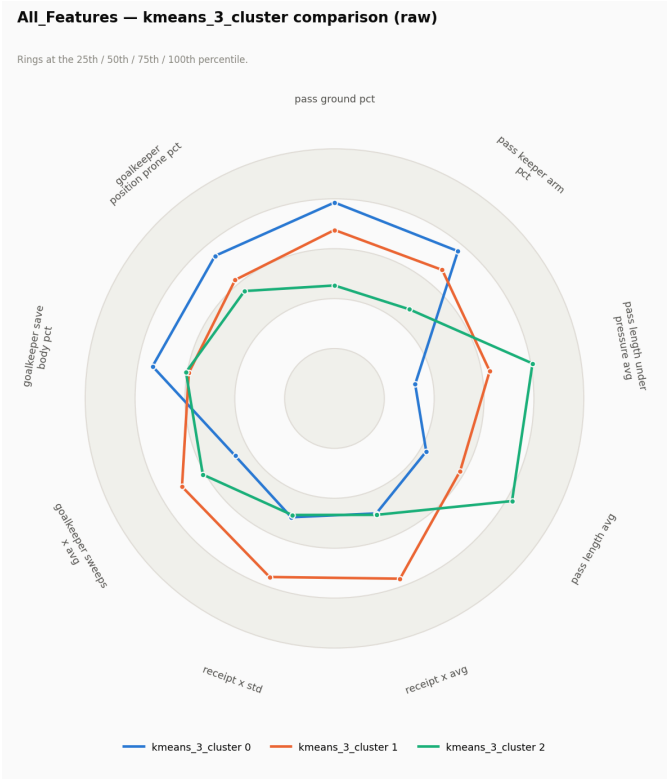


Figure 11: Raw Statistics Comparison

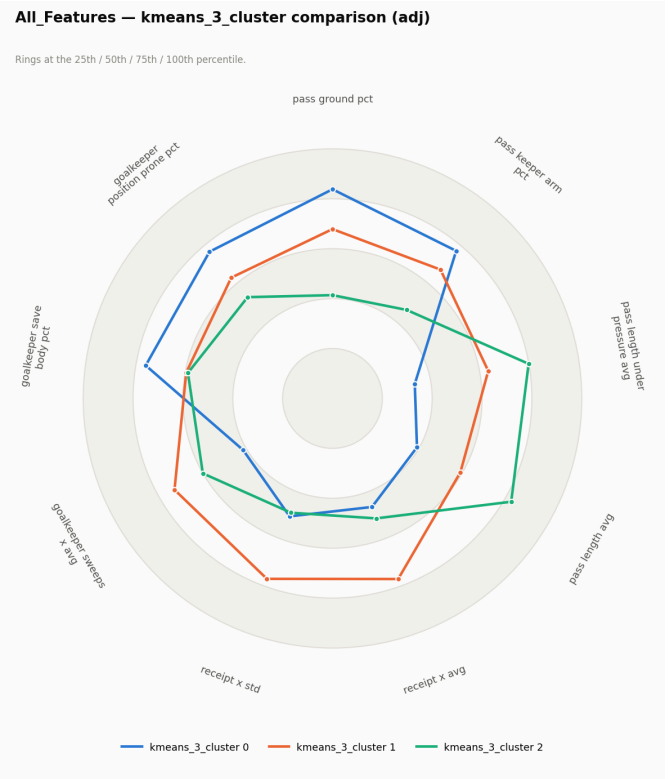


Figure 12: Adjusted Statistics Comparison

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FA Women's Super League

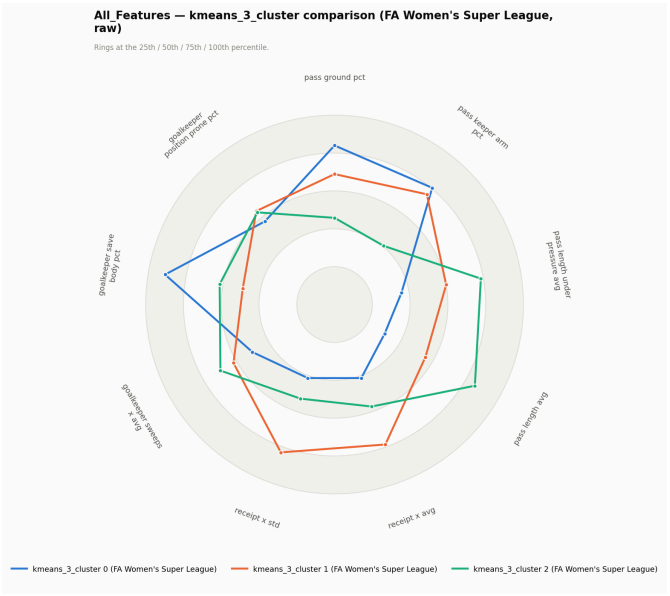


Figure 13: FA Women's Super League Goalkeepers — Raw Statistics Comparison

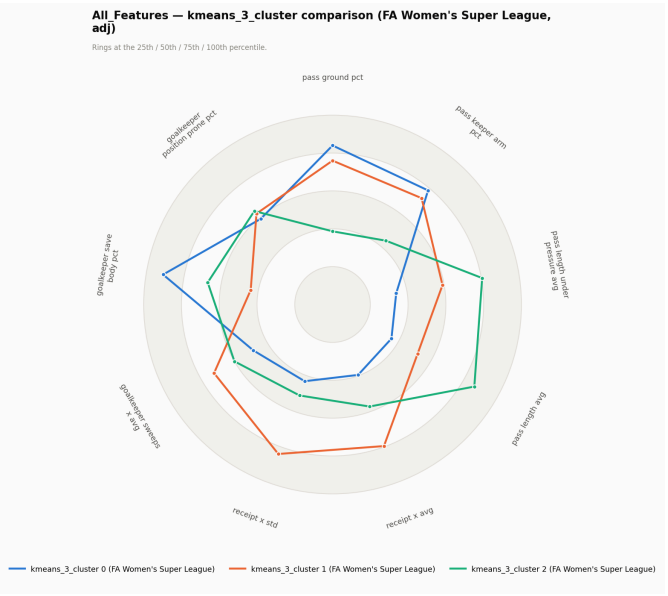


Figure 14: FA Women's Super League Goalkeepers — Adjusted Statistics Comparison

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Frauen Bundesliga

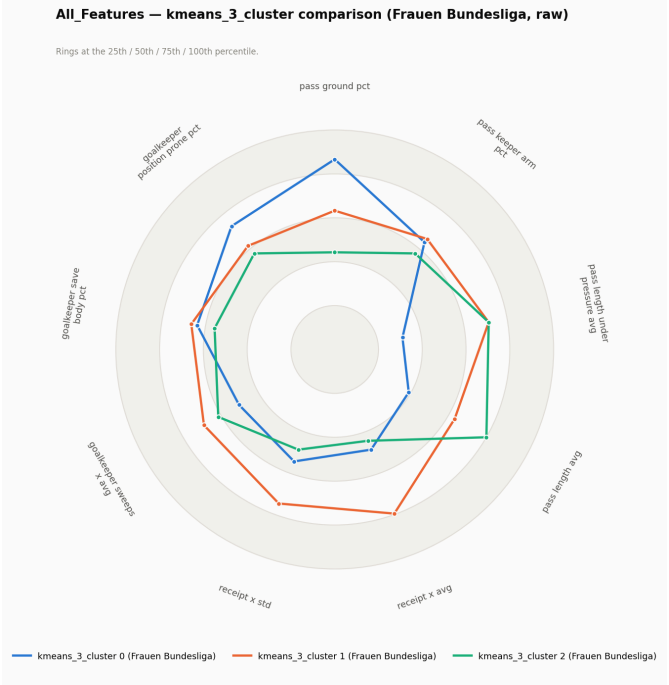


Figure 15: Frauen Bundesliga Goalkeepers — Raw Statistics Comparison

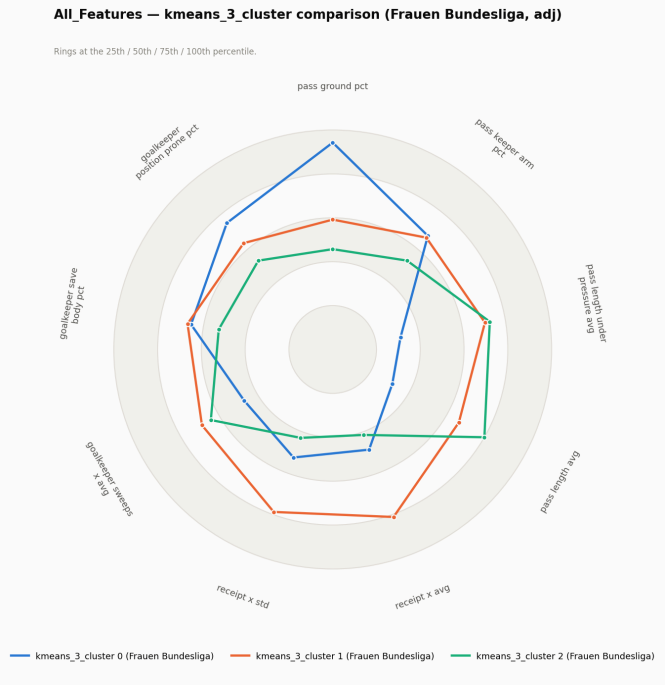


Figure 16: Frauen Bundesliga Goalkeepers — Adjusted Statistics Comparison

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Liga F

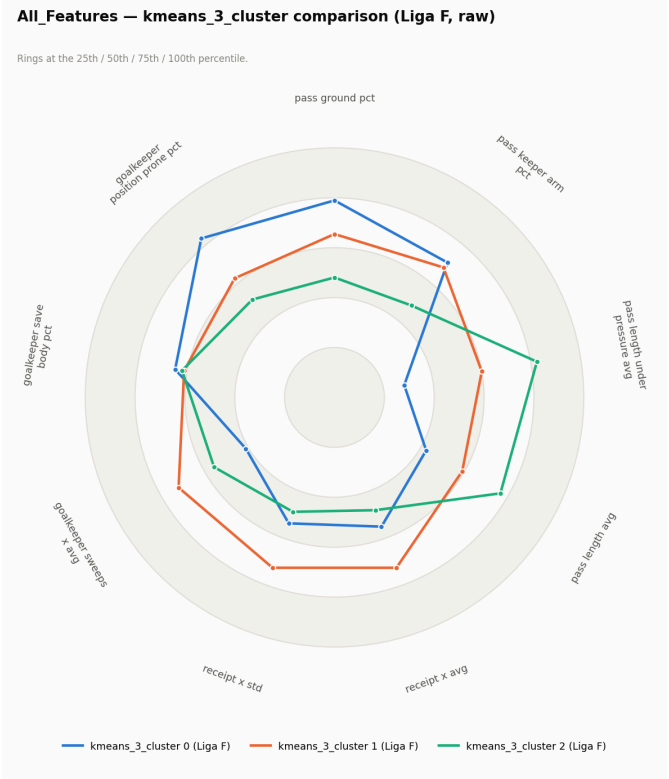


Figure 17: Liga F Goalkeepers — Raw Statistics Comparison

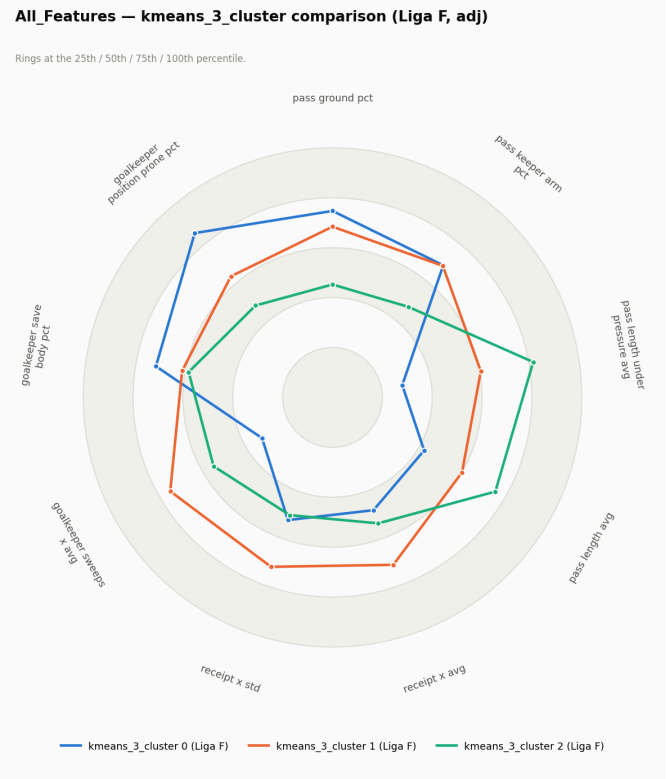


Figure 18: Liga F Goalkeepers — Adjusted Statistics Comparison

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Serie A Women

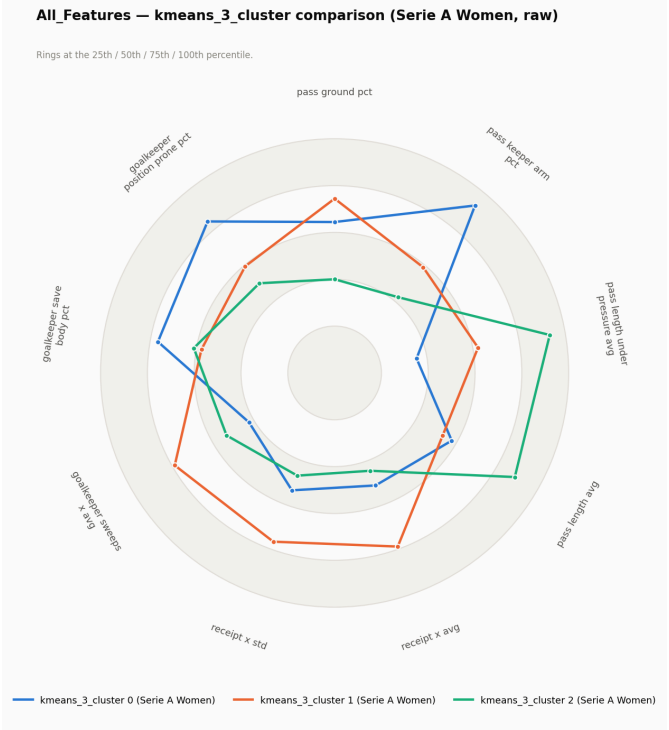


Figure 19: Serie A Women Goalkeepers — Raw Statistics Comparison

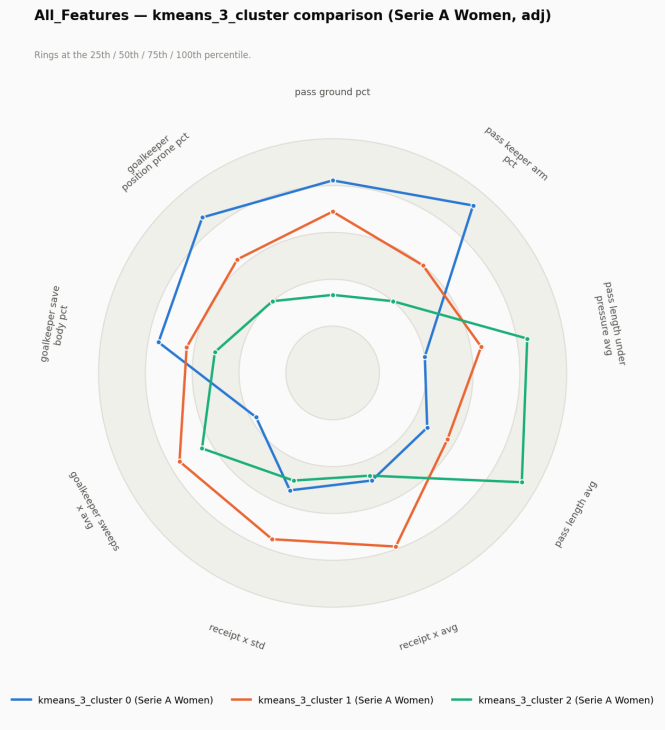


Figure 20: Serie A Women Goalkeepers — Adjusted Statistics Comparison